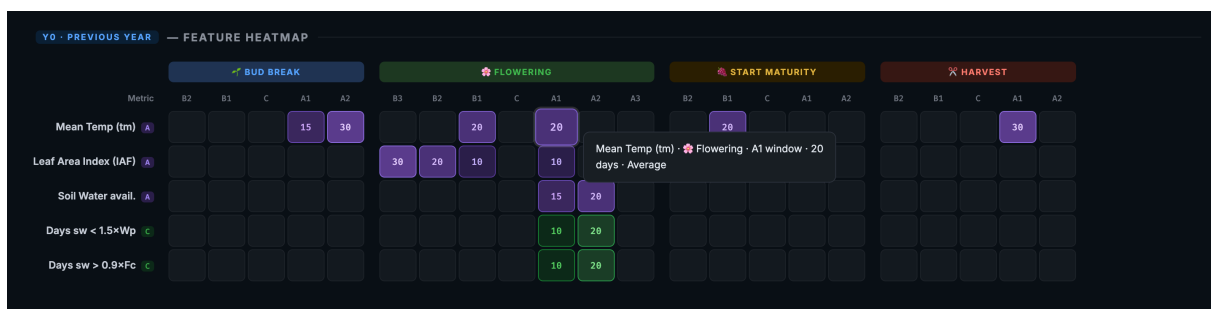
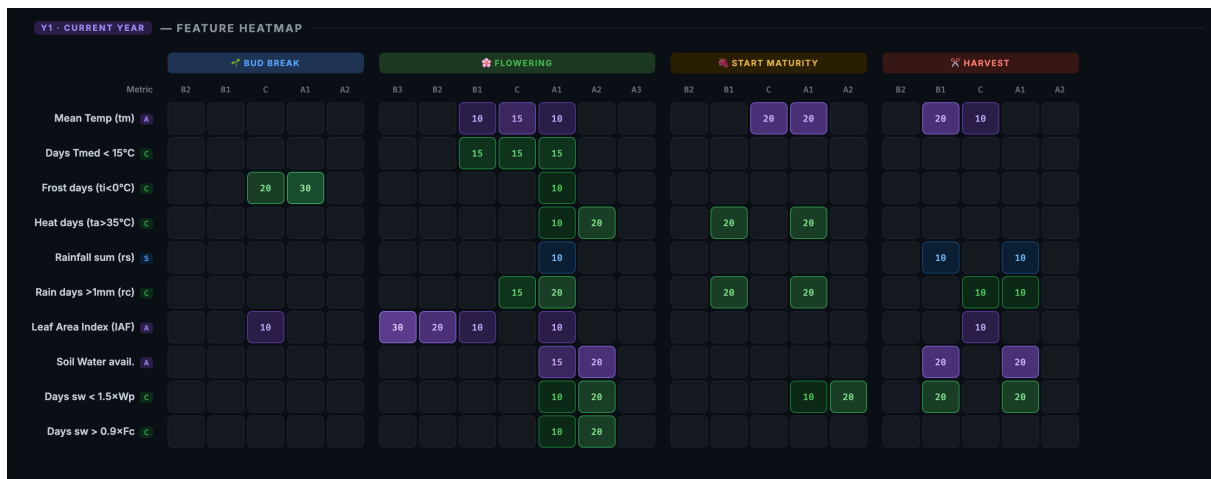
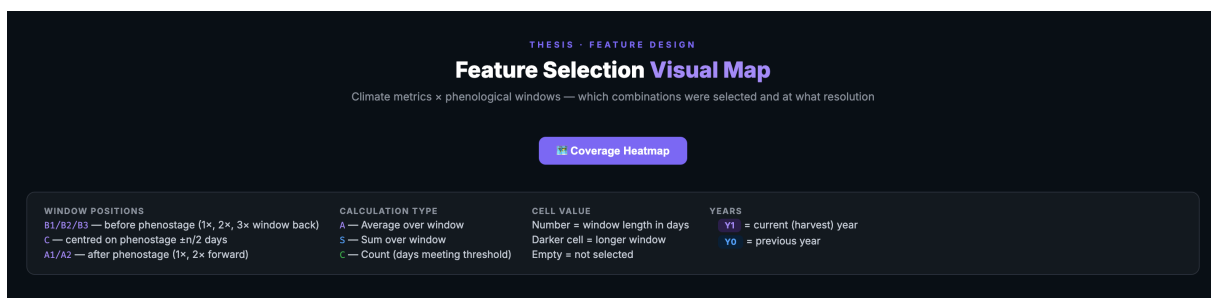




Features Database

⚙ Status	Done
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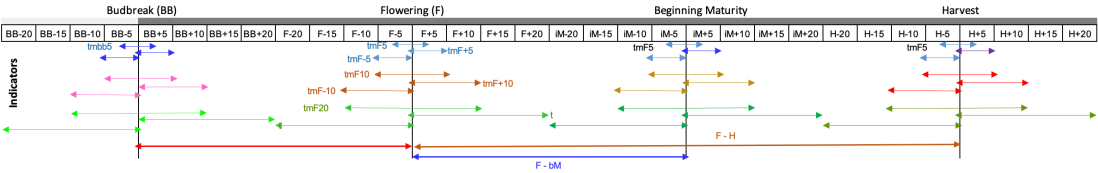
The lifecycle of grape development



VARIABLE REFERENCE TABLE						
SYMBOL	FULL NAME	CALC	UNIT	THRESHOLD	YEARS	DESCRIPTION
tm	Mean daily temperature	A Average	°C	—	Y1 Y0	Daily mean temperature averaged over the window. A core thermal proxy for vine phenology: temperatures drive bud burst timing, berry set, sugar accumulation, and ripening speed.
tm<15°C	Cold days (Tmed < 15°C)	C Count	days	Tmed < 15°C	Y1	Count of days where mean temperature drops below 15°C. During flowering, cold conditions below this threshold impair pollen viability and fertilisation, directly reducing berry set and yield potential.
ti<0°C	Frost days (Tmin < 0°C)	C Count	days	Tmin < 0°C	Y1	Count of days with minimum temperature below 0°C. Frost events around bud break can irreversibly damage newly emerged shoots, causing partial or total yield loss for that growing season.
ta>35°C	Heat stress days (Tmax > 35°C)	C Count	days	Tmax > 35°C	Y1	Count of days with maximum temperature exceeding 35°C. Extreme heat during flowering causes flower abortion and poor set; during maturation it accelerates sugar rise while impairing acid retention and aroma.
rs	Rainfall sum	S Sum	mm	—	Y1	Total accumulated rainfall over the window period. Captures the overall water input from precipitation — critical around flowering (dilution risk) and harvest (disease, berry splitting).
rc>1mm	Wet days (rain > 1mm)	C Count	days	Rain > 1mm	Y1	Count of days with rainfall exceeding 1mm. Frequent wet days increase humidity and disease pressure (downy mildew, botrytis), and during flowering they physically hinder pollination.
IAF	Leaf Area Index (IAF)	A Average	index	—	Y1 Y0	Índice de Área Foliar — solar radiation interception index averaged over the window. Reflects canopy size and photosynthetic capacity. High LAI before flowering indicates vigorous vegetative growth that can compete with reproductive development.
sw_a	Available soil water	A Average	mm	—	Y1 Y0	Mean soil water content available to the vine (between wilting point and field capacity). Sustained low values signal water stress; high values around harvest can reduce fruit concentration and quality.
<1.5×Wp	Severe water stress days	C Count	days	Sm < 1.5 × Wp	Y1 Y0	Count of days where soil water falls below 1.5× the wilting point — a threshold for severe vine water stress. Douro: Wp = 402 mm; Vinho Verde: Wp = 338 mm. Prolonged stress reduces yield but can concentrate phenolics.
>0.9×Fc	Soil saturation days	C Count	days	Sm > 0.9 × Fc	Y1 Y0	Count of days where soil water exceeds 90% of field capacity — indicating near-saturation conditions. Douro: Fc = 142.8 mm; Vinho Verde: Fc = 77 mm. Excess water limits root oxygen and increases disease risk.

PHENOLOGICAL STAGE	CALCULATION - WINDOW SIZE
BB Bud Break	10 Average – 10d
F1 Flowering	20 Average – 20d
SM Start of Maturity	30 Average – 30d
H Harvest	10 Count – 10d
	20 Count – 20d

▼ Example



This image illustrates the lifecycle of grape development, focusing on key phenological stages and their corresponding time intervals. Here is a detailed explanation of each section and the indicators used:

Lifecycle Stages

- Budbreak (BB):** The stage when buds begin to swell and burst open, initiating the growing season.
- Flowering (F):** The period when grapevine flowers open and pollination occurs.
- Beginning Maturity (iM):** The stage marking the start of grape maturation, when berries begin to soften and change color.

4. **Harvest (H):** The final stage when grapes are fully mature and ready for picking.

Time Intervals

Each phenological stage is marked by a central point (e.g., BB for Budbreak, F for Flowering), and time intervals are defined relative to these points:

Negative Values: Days before the phenological event (e.g., BB-20 means 20 days before Budbreak).

Positive Values: Days after the phenological event (e.g., BB+20 means 20 days after Budbreak).

Indicators and Arrows

The diagram uses arrows to indicate various meteorological and phenological measurements taken during these intervals:

Blue Arrows: Represent measurements taken over a shorter period around the central phenological date (e.g., tmBB5, tmF5), usually 5 days before and after the event.

Pink Arrows: Show measurements taken for slightly longer periods before and after the phenological date (e.g., 10 or 15 days).

Brown Arrows: Indicate measurements taken over even longer periods, such as 20 days before and after the event.

Green Arrows: Represent measurements taken over extended periods that span across multiple stages (e.g., from Budbreak to Flowering, Flowering to Harvest).

Red Arrows: Often show measurements or indicators taken at the end stages, like near Harvest.

Example Metrics

tmBB5: Mean temperature over a 5-day period centered around Budbreak (BB-5 to BB+5).

tmF5: Mean temperature over a 5-day period centered around Flowering (F-5 to F+5).

tmF10: Mean temperature over a 10-day period centered around Flowering (F-10 to F+10).

tmF20: Mean temperature over a 20-day period centered around Flowering (F-20 to F+20).

Detailed Intervals

1. Budbreak (BB):

BB-20 to BB+20: Indicates a total period spanning 40 days around Budbreak.

Shorter intervals like **BB-5 to BB+5**, **BB-10 to BB+10**, and **BB-15 to BB+15** are also shown.

2. Flowering (F):

F-20 to F+20: Indicates a total period spanning 40 days around Flowering.

Similar shorter intervals are indicated around Flowering as well.

3. Beginning Maturity (iM):

iM-20 to iM+20: Indicates a total period spanning 40 days around the Beginning of Maturity.

Includes shorter intervals like **iM-5 to iM+5**, **iM-10 to iM+10**, etc.

4. Harvest (H):

H-20 to H+20: Indicates a total period spanning 40 days around Harvest.

Includes shorter intervals similar to other stages.

▼ Features creation

Grapevine Phenological Stages and Climate Indicators

Mean Temperature (tm)

Flowering (F)

- Current Year (Y1)

- Average over 10 days before Flowering - **F-10** (tm_fl_y1_b10)
- Average over 15 days centered around Flowering - **F-7 to F+7** (tm_fl_y1_c15)
- Average over 10 days after Flowering - **F+10** (tm_fl_y1_a10)
- Count number of days with Mean Temperature below 15°C over 15 days centered around Flowering - **F-7 to F+7** (tm.days.less.15_fl_y1_c15)
- Count number of days with Mean Temperature below 15°C over 15 days before Flowering - **F-15** (tm.days.less.15_fl_y1_b15)
- **Previous Year (Y0)**
 - Average over 20 days before Flowering - **F-20** (tm_fl_y0_b20)
 - Average over 20 days after Flowering - **F+20** (tm_fl_y0_a20)

Bud Break (BB)

- **Previous Year (Y0)**
 - Average over 15 days after Bud Break - **BB+15** (tm_bb_y0_a15)
 - Average over 30 days after Bud Break - **BB+30** (tm_bb_y0_a30)

Start of Maturity (sM)

- **Current Year (Y1)**
 - Average over 20 days centered around Start of Maturity - **sM-10 to sM+10** (tm_sm_y1_c20)
 - Average over 20 days after Start of Maturity - **sM+20** (tm_sm_y1_a20)
- **Previous Year (Y0)**
 - Average over 20 days before Start of Maturity - **sM-20** (tm_sm_y0_b20)

Harvesting (H)

- **Current Year (Y1)**
 - Average over 10 days centered around Harvesting - **H-5 to H+5** (tm_hv_y1_c10)

- Average over 20 days before Harvesting - **H-20** (tm_hv_y1_b20)
- **Previous Year (Y0)**
 - Average over 30 days after Harvesting - **H+30** (tm_hv_y0_a30)

Minimum Temperature < 0°C (ti)

Bud Break (BB)

- **Current Year (Y1)**
 - Count number of days with Minimum Temperature below 0°C over 20 days centered around Bud Break - **BB-10 to BB+10** (days_minTemp.less.0_bb_y1_c20)
 - Count number of days with Minimum Temperature below 0°C over 30 days after Bud Break - **BB+30** (days_minTemp.less.0_bb_y1_a30)

Flowering (F)

- **Current Year (Y1)**
 - Count number of days with Minimum Temperature below 0°C over 10 days after Flowering - **F+10** (days_minTemp.less.0_fl_y1_a10)

Maximum Temperature > 35°C (ti)

Flowering (F)

- **Current Year (Y1)**
 - Count number of days with Maximum Temperature above 35°C over 10 days after Flowering - **F+10** (days_maxTemp.above.35_fl_y1_a10)
 - Count number of days with Maximum Temperature above 35°C over 20 days after Flowering - **F+20** (days_maxTemp.above.35_fl_y1_a20)

Start of Maturity (sM)

- **Current Year (Y1)**
 - Count number of days with Maximum Temperature above 35°C over 20 days before Start of Maturity - **sM-20** (days_tempMax.above.35_sm_y1_b20)

- Count number of days with Maximum Temperature above 35°C over 20 days after Start of Maturity - **sM+20**
(days.maxTemp.above.35_sm_y1_a20)

Rainfall (rf)

Flowering (F)

- **Current Year (Y1)**
 - Sum of rain quantity over 10 days after Flowering - **F+10**
(rf_fl_y1_a10)
 - Count number of days with rain >1mm over 15 days centered around Flowering - **F-7 to F+7** (days.rf.above.1mm_fl_y1_c15)
 - Count number of days with rain >1mm over 20 days after Flowering - **F+20** (days.rf.above.1mm_fl_y1_a20)

Harvesting (H)

- **Current Year (Y1)**
 - Sum of rain quantity over 10 days before Harvesting - **H-10**
(rf_hv_y1_b10)
 - Sum of rain quantity over 10 days after Harvesting - **H+10**
(rf_hv_y1_a10)
 - Count number of days with rain >1mm over 10 days centered around Harvesting - **H-5 to H+5** (days.rf.above.1mm_hv_y1_c10)
 - Count number of days with rain >1mm over 10 days after Harvesting - **H+10** (days.rf.above.1mm_hv_y1_a10)

Indice Area Foliar (IAF)

Bud Break (BB)

- **Current Year (Y1)**
 - Average values over 10 days centered around Bud Break - **BB-5 to BB+5** (iaf_bb_y1_c10)

Flowering (F)

- **Current Year (Y1)**

- Average values over 10 days before Flowering - **F-10** (iaf_fl_y1_b10)
- Average values over 20 days before Flowering - **F-20** (iaf_fl_y1_b20)
- Average values over 30 days before Flowering - **F-30** (iaf_fl_y1_b30)
- Average values over 10 days after Flowering - **F+10** (iaf_fl_y1_a10)

- **Previous Year (Y0)**

- Average values over 10 days before Flowering - **F-10** (iaf_fl_y0_b10)
- Average values over 20 days before Flowering - **F-20** (iaf_fl_y0_b20)
- Average values over 30 days before Flowering - **F-30** (iaf_fl_y0_b30)
- Average values over 10 days after Flowering - **F+10** (iaf_fl_y0_a10)

Harvesting (H)

- **Current Year (Y1)**

- Average values over 10 days centered around Harvesting - **H-5 to H+5** (iaf_hv_y1_c10)

Available Soil Water (SWA)

Flowering (F)

- **Current Year (Y1)**

- Average values over 15 days after Flowering - **F+15** (swa_fl_y1_a15)
- Average values over 30 days after Flowering - **F+30** (swa_fl_y1_a30)
- Count number of occurrences where soil water is $< 1.5 \times Wp$ over 10 days after Flowering - **F+10** (sw.less.15wp_fl_y1_a10)
- Count number of occurrences where soil water is $< 1.5 \times Wp$ over 20 days after Flowering - **F+20** (sw.less.15wp_fl_y1_a20)
- Count number of occurrences where soil water is $> 0.9 \times Fc$ over 10 days after Flowering - **F+10** (sw.above.09fc_fl_y1_a10)
- Count number of occurrences where soil water is $> 0.9 \times Fc$ over 20 days after Flowering - **F+20** (sw.above.09fc_fl_y1_a20)

- **Previous Year (Y0)**

- Count number of occurrences where soil water is $< 1.5 \times Wp$ over 10 days after Flowering - **F+10** (sw.less.15wp_fl_y0_a10)
- Count number of occurrences where soil water is $< 1.5 \times Wp$ over 20 days after Flowering - **F+20** (sw.less.15wp_fl_y0_a20)
- Count number of occurrences where soil water is $> 0.9 \times Fc$ over 10 days after Flowering - **F+10** (sw.above.09fc_fl_y0_a10)
- Count number of occurrences where soil water is $> 0.9 \times Fc$ over 20 days after Flowering - **F+20** (sw.above.09fc_fl_y0_a20)

Start of Maturity (sM)

- **Current Year (Y1)**
 - Count number of occurrences where soil water is $< 1.5 \times Wp$ over 10 days after Start of Maturity - **sM+10** (sw.less.15wp_sm_y1_a10)
 - Count number of occurrences where soil water is $< 1.5 \times Wp$ over 20 days after Start of Maturity - **sM+20** (sw.less.15wp_sm_y1_a20)

Harvesting (H)

- **Current Year (Y1)**
- Average values over 20 days before Harvesting - **H-20** (swa_hv_y1_b20)
- Average values over 20 days after Harvesting - **H+20** (swa_hv_y1_a20)
- Count number of occurrences where soil water is $< 1.5 \times Wp$ over 20 days before Harvesting - **H-20** (sw.less.15wp_hv_y1_b20)
- Count number of occurrences where soil water is $< 1.5 \times Wp$ over 20 days after Harvesting - **H+20** (sw.less.15wp_hv_y1_a20)